Quiz #6 Review

1 a state the p-series test

- $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p > 1$ & divergent if $p \leq 1$

b) State the test for divergence

If $\lim_{n \to \infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is divergent

c) State the integral test

Let f be a continuous, positive, decreasing function on $[1, \infty)$ and let $a_n = f(n)$

- If $\int_1^{\infty} f(x)$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent

If $\int_1^{\infty} f(x)$ is divergent, then $\sum_{n=1}^{\infty} a_n$ is divergent.

d) State the geometric series test.

$\sum_{n=1}^{\infty} ar^{n-1}$ is convergent if, $|r| < 1$

is divergent if $|r| \geq 1$

2 Find the limits of the sequences.
If bounded, show/give the bounds.
Determine if its increasing, decreasing, or not monotonic.

a) $a_n = \frac{2^n}{n^2}$, $\{2, 1, 8/9 \ldots\}$

$\lim_{n\to\infty} \frac{2^n}{n^2} = \infty$

not monotonic
not bounded

b) $a_n = \{ 1/2, \frac{4}{6}, \frac{7}{10}, \frac{10}{14} \ldots \}$

$a_n = \frac{3n-2}{4n-2}$, $\lim_{n\to\infty} \frac{3n-2}{4n-2} = 3/4 \rightarrow$ upperbound

monotonically increasing
$1/2 =$ lower bound.

3 Determine whether the series is convergent or divergent

a)

$$\sum_{k=1}^{\infty} \frac{1}{k^2+4k+3}$$

$$\frac{1}{k^2+4k+3} = \frac{1}{(k+1)(k+3)} = \frac{A}{k+1} + \frac{B}{k+3}$$

$$A(k+3) + B(k+1) = 1$$

$$\Rightarrow \quad Ak + Bk + 3A + B = 1$$

$$3A + B = 1$$

$$A + B = 0$$

$$A = 1/2, \quad B = -1/2$$

$$\sum_{k=1}^{\infty} \frac{1}{k^2+4k+3} = \sum_{k=1}^{\infty} \frac{1}{2}/(k+1) - 1/2/(k+3)$$

$$= \left(\frac{1/2}{2} + \frac{1/2}{3} + \cancel{\frac{1/2}{4} \cdots}\right) - \left(\cancel{1/2/4 + \frac{1}{2}/5 + \cdots}\right)$$

$$= \frac{1/2}{2} + \frac{1/2}{3} = 5/12$$

b) $\sum_{k=1}^{\infty} \frac{e^{k+2}}{3^{k+1}}$

$= \sum_{k=1}^{\infty} \frac{e^2}{3} \cdot \frac{e^k}{3^k}$

$= \sum \frac{e^2}{3} \cdot \underbrace{\left(\frac{e}{3}\right)}_{r}^k$

$|r| < 1$ hence, by the Geometrix series test, the sum converges.

4) Express $0.32\overline{32}$ as a ratio of integers using finite series. show work.

$= \sum_{n=0}^{\infty} \frac{32}{100}\left(\frac{1}{100}\right)^n$

$\text{Sum} = \frac{a}{1-r} = \frac{\frac{32}{100}}{1-\frac{1}{100}} = \boxed{\frac{32}{99}}$

5 a) Estimate the Error Using the partial sum S_{20} of the series $\sum_{k=1}^{\infty} \frac{1}{k^4}$

$$\lim_{t \to \infty} \int_{20}^{t} k^{-4}\, dk = \lim_{t \to \infty} \frac{-1}{3k^3}\Big|_{20}^{t}$$

$$= \lim_{t \to \infty} \frac{-1}{3t^3} + \frac{1}{3(20)^3} = \frac{1}{3(20)^3}$$

$$\int_{21} f(x)\, dx \leq R_{20} \leq \int_{20}^{\infty} f(x)\, dx$$

hence

$$\frac{1}{3(21)^3} \leq R_{20} \leq \frac{1}{3(20)^3}$$

b) What partial sum would be needed to han an accuracy within 10^{-4}?

$$\frac{1}{3n^3} < \frac{1}{10^4}, \quad n > \sqrt[3]{\frac{10^4}{3}}.$$

next integer